

GOUTHAMDAS K

Electronics and Biomedical Engineer

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📍 Kannur, Kerala

SUMMARY

Motivated B.Tech graduate in Electronics and Biomedical Engineering with a strong interest in medical equipment technology, device servicing, and embedded healthcare systems. Proven ability in designing and prototyping biomedical solutions, including a multi-sensor stress-monitoring wearable. Hands-on exposure to critical-care and diagnostic equipment through internships and industry training. Brings strong analytical thinking, troubleshooting skills, and professional communication suited for roles in biomedical service engineering, technical support, and product deployment.

EXPERIENCE

Sales & Service Intern

Biolab IVD

📅 01/02/2026 – Present 📍 Kannur, Kerala

A company specializing in in-vitro diagnostic (IVD) laboratory equipment and technical support services.

- Assisted in the installation, commissioning, and calibration of biochemistry and hematology analyzers at diagnostic laboratories and hospitals
- Performed preventive maintenance and troubleshooting of laboratory diagnostic equipment under senior service engineers
- Supported customer training on equipment operation, routine maintenance, and quality control procedures
- Conducted fault diagnosis and component replacement to minimize equipment downtime
- Maintained service reports, installation records, and technical documentation in compliance with company procedures
- Gained hands-on experience in field service operations, customer support, and laboratory workflow management

Biomedical Intern

Baby Memorial Hospital

📅 09/06/2025 – 20/06/2025 (10 days) 📍 Kannur, Kerala

A hospital providing comprehensive healthcare services

- Studied installation, testing and functioning of medical devices across ICU, OT, Radiology, Laboratory
- Learned calibration workflow & preventive maintenance for critical-care equipment
- Understood troubleshooting procedures under senior biomedical engineers
- Observed radiology systems: X-ray, CT, MRI, Ultrasound

Industrial Training

Skill Biomed Industrial Training Programme

📅 01/08/2022 – 31/08/2022 (1 Month) 📍 Ernakulam, Kerala

- Hands-on training supported by AKAS infusions, Confidential Dental & Erbe Medical.
- Covered major devices: Patient Monitor, Fetal Monitor, Defibrillator, AED, ESU, Oxygen Concentrator, ECG.
- Lab systems: Urine Analyzer, Biochemistry Analyzer, Hematology Analyzer.
- Dental devices: Dental X-ray, RVG, PSP, Autoclave, Dental Chair.
- Critical-care: Syringe Pump, Infusion Pump, ICU Ventilator, Electrical Safety Analyzer, Multi-para Simulator.

Biomedical Intern

AKG Memorial Hospital

📅 14/10/2021 – 20/10/2021 (7 days) 📍 Kannur, Kerala

A hospital providing comprehensive healthcare services

- Studied workflows in ICU, Radiology & Laboratory departments.
- Exposure to equipment handling, maintenance & troubleshooting culture.
- Certificate notes sincerity, efficiency, and competence.

EDUCATION

B Tech – Electronics & Biomedical Engineering | APJKTU

Adi Shankara Institute of Engineering and Technology

Ernakulam, Kerala, India

Sep 2023 – May 2026

Diploma – Biomedical Engineering | State Board of Technical Education

EKNM Model Polytechnic College

Kannur, Kerala, India

Aug 2020 – Apr 2022

Plus Two (HSC) – Science | Board of HSE, Govt of Kerala

Parassinikadavu Higher Secondary School

Kannur, Kerala, India

July 2017 – April 2019

PROFESSIONAL SKILL

- Medical Equipment Exposure (ICU/OT/Radiology/Lab)
- Basic device testing & validation
- Documentation and reporting
- PCB Soldering & basic circuit build
- Embedded prototyping
- Problem Solving
- Adaptability
- MATLAB basic Modeling
- MS Office
- Embedded C fundamentals
- KiCAD
- Analytical Reasoning
- Teamwork
- Technical Communication

PROJECT

Real-Time Stress Detection & Management System (2025)

Developed a wearable biomedical monitoring prototype integrating multiple physiological sensors for stress assessment.

Key Contributions:

- Interfaced MAX30102, ECG, GSR, FSR402 and DHT22 sensors for real-time data acquisition
- Designed ESP32-based embedded system for signal processing and basic stress detection logic
- Implemented alert outputs using GSM module, OLED display, vibration actuator and audio indication
- Designed Li-ion rechargeable power architecture using TP4056 and XL6009
- Demonstrated functional prototype suitable for health monitoring applications

ACHIEVEMENTS/WORKSHOPS

- **Biomedical Equipment & Safety Standards Workshop:** Completed a 5-day hands-on biomedical engineering workshop (21–25 Oct 2025) conducted by i-CET (Inskon Medical Solutions Pvt. Ltd.) in association with the **Telemedicine Society of India** at ASIET, Kalady. Covered ventilators, anesthesia workstations, electrosurgical units, drug infusion systems, and medical device safety standards including calibration, validation, and electrical safety testing. (2025)
- **Conference Presentation:** Presented research work titled “Real-Time Stress Detection and Management System with Feedback Mechanism” at the International Conference on Innovations in Mechanical, Robotics, Computing and Biomedical Engineering (ICIMRBE 2025), organized by ASIET in collaboration with A2Z EduScaleUp India Pvt Ltd (2025)
- Participated in **HOSPEX Healthcare Expo 2025** workshop on Healthcare Procurement and Supply Chain Practices, held at KINFRA International Exhibition Centre, Kochi (2025).
- Actively participated in **Ultrasound Workshop and Biomedical Equipment Workshop**, hosted by Biomedical Association (BMA) of Govt. Model Engineering College in collaboration with CYRIX Healthcare as part of EXCEL 2025
- Attended **IVD Instrumentation Workshop** conducted by Horiba India Technical Institute (2024).
- Participated in **Arduino In Action Hands-on Workshop** organized by IEDC RIT (2024).
- Successfully completed **Electronic Product Building Workshop**, conducted by Truetek World in association with EBSAA and BMESI (2023).

LANGUAGES

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| • English | • Hindi | • Malayalam | • Tamil |
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